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ORIGINAL ARTICLE

A dual-stage control system for high-speed, ultra-precise linear motion

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Abstract Modern manufacturing equipment often requires high-speed and ultra-precise linear motion. For implementing such motion, a coarse–fine dual stage is effective because the coarse stage has a low bandwidth with a large workspace, and in contrast, the fine stage has a high bandwidth with a small stroke. This paper presents the implementation of an air-bearing, dual-stage system and its control strategy. A closed-loop feedback in an absolute space is used to realize coarse–fine control. The fine stage is driven by a voice coil motor that tracks the designed trajectory, while the coarse stage is driven by permanent-magnet linear synchronous motor that tracks the trajectory of the fine stage to prevent its motion saturation. Also analyzed are the coupled dynamics of the air-bearing dual stage driven by a direct-drive motor. Identification and robust control design for the fine stage are introduced in detail. Method tests for the single fine stage are performed, and the results demonstrate that ultra-precision control can be realized for the fine stage. Next, experiments are presented with the dual stage using different loads. Experimental results show that our control strategy can achieve high-speed, ultra-precision linear motion through the dual stage with a satisfactory performance.

Keywords Control system · Linear motion platform · Coarse–fine control · Model identification

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1 Introduction

A high-speed, ultra-precise control system over a large workspace is an essential technology in modern manufacturing, for example, in precision optics alignment, microbiological cell manipulation, and semiconductor manufacturing fields. Several compound actuation systems with dual-stage construction were investigated for different applications. These served as trial methods for overcoming the limitations of conventional actuators. The purpose of compound actuation is to achieve both of the capacities provided by the coarse and fine actuators in a system using their respective merits.

In a compound actuation system, the coarse and fine actuators complement each another. In other words, the long reach, which is the coarse manipulator characterized by a "slow" response due to its size, offers a large workspace, while the short reach, which is the fine manipulator characterized by a small work volume with a fast, precise motion capability over that work volume, enables a high resolution of motion. Combining these two, so that the fine manipulator rides on the coarse manipulator, offers a solution to the requirement for a fast, precise motion over a large workspace. As such, any defects of one actuator can be compensated for by the merits of the other actuator.

There have been reports in published literature about dual-stage linear motion control systems. As a coarse servo, several actuators, such as a DC motor, ball screw, voice coil motor, and linear motor, can be used. As a fine servo, Piezoelectric Transducer (PZT) is frequently used to provide high-precision positioning, with attractive features such as very high positioning resolution, high stiffness, high output force, and medium cost, but its positioning ranges only in micrometers or several hundreds of microns at the most. The PZT's application area is restricted to micro-motion due to its limited displacement. Considering this factor, the tracking capability degradation caused by motion saturation should be prevented, and therefore, the error level in the coarse stage should be as small as possible. For example, the dynamically consistent theorem [1] in null space to prevent motion saturation of the linear dual stages [2] and a perturbation compensator was designed to improve the performance of the coarse stage [3].

Recent works has focused on putting more effort on increasing the movement range of the PZT. The linear piezomotors employing three to five PZT are common, but their performance varies with individual designs, and their magnified hysteresis characteristics and flexibility lead to a serious decrease of the control bandwidth and an increase of the controller design's complexity for achieving ultra-precision dynamic response, especially with a heavy payload. In effect, the PZT has the advantage of accomplishing positioning tasks rather than ultra-precision tracking tasks with a heavy payload. A magnetostrictive actuator with a bandwidth in the kilohertz range capable of several kilonewtons of force output can be used as the fine positioning element for a cutting machine [4]. Like a PZT, it displays a significant hysteresis and makes the effective implementation of real-time fast servo control challenging.

The position of a coarse–fine (or macro–micro) actuator can be controlled either sequentially or simultaneously. In some applications, fine and coarse positioning are separated and repeated in a sequential manner. For a high-speed task, positioning or tracking control for the fine stage is usually accomplished during the control of the coarse stage. However, some approaches of the coarse–fine simultaneous controller can be employed by different mechanisms. These approaches are usually classified into multi-input multi-output (MIMO) and single-input single-output (SISO) designs, depending on the plant dynamics.

In almost all of the literature, the linear dual stage can be considered as a SISO system under the assumption that there is little interaction between two actuator systems. On the premise of a definite dual-stage precision level and the response speeds of the two subsystems separated from each other, such a dynamic coupling can be neglected. This can only be done if the fine actuator has enough closed loop stiffness to reject disturbances due to coupling, and the bandwidth of the fine stage is below the fundamental resonant frequency of the coarse actuator. With a heavy load, the bandwidths of the coarse and fine stages are comparatively close. However, the coupling should be carefully evaluated by various types of coarse–fine mechanisms.

The position output approaches of the dual-stage mechanism in absolute space are also related to the coupling analysis. In some coarse–fine stages, the servo reference for the fine actuator is to track the coarse stage's error through closed loop feedback control using a built-in relative position sensor (i.e., the end-point position of the dual-stage mechanism is equal to the sum of two motions) [3]. In this case, the inertia force acting on the fine stage due to the coarse motion should be considered in ultra-precision control, even if there are no friction and damp between the two subsystems. The accuracy of the dual stages, equipped only with built-in sensors, may be worse than that of the single stage because of installation errors. Moreover, stricter mechanical calibrations through external measurement are required to compensate for such mounting errors. A straightforward way to increase the precision of the positioning and tracking tasks is to directly measure the end-point position using a sensor mounted on the fine stage. The performance of a coarse–fine actuator was proven to substantially improve by implementing a coarse–fine control algorithm based on absolute position sensing. An end-point sensor might be the optimal solution to achieve the highest possible accuracy and repeatability. This end-point position control strategy provides a convenient and effective way to design a controller in absolute space.

We have constructed an air-bearing, dual-stage servo system that provides a nanometer-level tracking error and a long range for high-speed travel with a heavy load. The coarse stage is driven by permanent-magnet linear synchronous motors (PMLSM). Air bearing is also adopted for the coarse stage to quicken its response speed and prevent motion saturation for the fine stage. With micron-level positioning and steady tracking of the coarse stage, a tracking capability ten times more effective than that of the coarse stage should be achieved. The fine stage is driven by voice coil motor (VCM), which has a longer stroke and a better dynamic response under a heavy load.

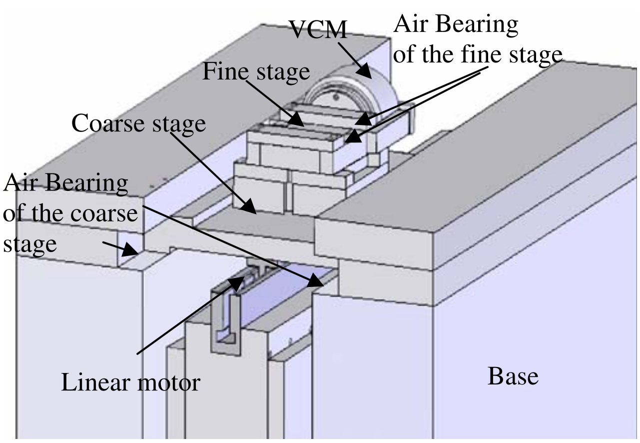
In our controller design, ultra-precise control for the fine stage is essential to ensure dual-stage accuracy. Meanwhile, the robustness of the controller is required because of an external disturbance, such as cable disturbances. Obtaining the model of the fine stage through identification methods is useful for the controller design and robustness estimation. The air-bearing stage provides a better dynamic response due to its low damp characteristic. However, this characteristic promotes inconsistency of the model identification because the system is unstable under an open-loop mode.

In our control system, we adopted a SISO design. Thus, the dynamic interaction of the dual stage is analyzed after the mathematical structure is introduced. Then, the identification and control algorithms for the fine stage are introduced in detail, and method tests for the single fine stage are performed (robustness analysis, positioning precision, and so on). Next, the dual-stage experimental setup is presented. As vibration may worsen the tracking performance, attention is given to the reference trajectory generation in Section 4. Finally, trajectory tracking results of the coarse–fine control system and conclusions are given.

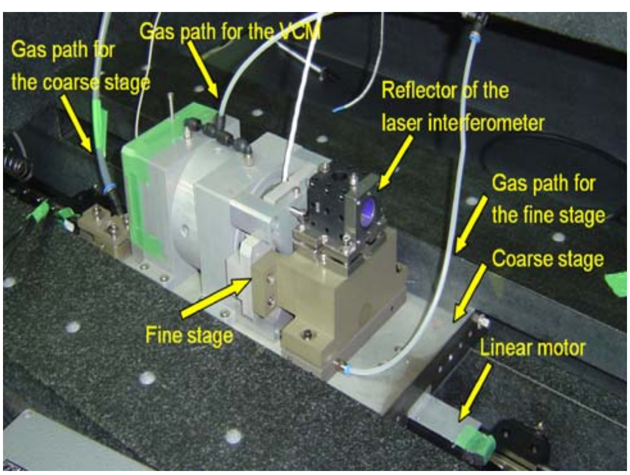
2 Description of the dual-stage control system

2.1 Mechanical structure

Traditional precise stages are usually actuated by ball screws coupled to a rotary servomotor, but there are many faults in the mechanism such as friction, backlash, and gear reduction. In addition, the closed loop bandwidth of the system will be limited by these nonlinear factors. In order to eliminate the aforementioned faults, an air-bearing dual stage driven by a direct-drive motor is adopted in our experimental dual-stage system, as shown in Fig. 1. The fine stage is driven by VCM with a maximum electromagnetic force of 334 N and a stroke of 10 mm. The coarse stage is driven by PMLSM, and its workspace is about 300 mm. For a precise control system for such a complicated mechanism, a non-collocated actuator and sensor may degrade control performance due to their structure flexibility. Therefore, an exposed linear encoder with a resolution of 0.1 μm is mounted on the surface of a granite bench and serves as the position feedback sensor for the coarse stage. The endpoint of the dual stage in absolute space is measured by a dual frequency laser interferometer with a 1.25-nm resolution and a 5 M/s maximum sampling rate. The reflector of the laser interferometer is mounted on the fine stage.



(a) The sketch of the mechanical structure of the dual stage



(b) Photograph of the experimental dual stage

Fig. 1 Mechanical structure of the dual stage

2.2 Dual control structure and analysis of dynamic coupling between two subsystems

The coarse–fine system transfer matrix in a frequency domain, which is currently common in industry, can be described as:

$$\begin{pmatrix} Y_c \\ Y_f \end{pmatrix} = \begin{pmatrix} \frac{M_e\left(M_2s^2+Bs+K\right)}{M_1M_2s^2(M_es^2+Bs+K)} & \frac{M_e-M_2}{M_2(M_es^2+Bs+K)} \\ \frac{M_e(Bs+K)}{M_1M_2s^2(M_es^2+Bs+K)} & \frac{M_e}{M_2(M_es^2+Bs+K)} \end{pmatrix} \begin{pmatrix} U_c \\ U_f \end{pmatrix} \quad (1)$$

where M_1 and M_2 are equivalent mass of the load of the coarse and fine stages, respectively; K and B are coupling spring constant and viscous damping coefficient, respectively; U_c, U_f are output forces of coarse and fine controller, respectively; Y_c represents output position of the coarse stage, and Y_f is output position of fine stage in absolute space. $M_e = \frac{M_1M_2}{M_1+M_2}$ is the equivalent mass for the coupling resonance.

For an air-bearing table, which provides zero friction, B and K can be ignored. Equation 1 then becomes:

$$\begin{pmatrix} Y_c \\ Y_f \end{pmatrix} = \begin{pmatrix} \frac{1}{M_1s^2} & \frac{-1}{M_1s^2} \\ 0 & \frac{1}{M_2s^2} \end{pmatrix} \begin{pmatrix} U_c \\ U_f \end{pmatrix} \quad (2)$$

Figure 2 shows the block diagram of the dual-stage control system, where r is the reference trajectory and d_f is the force disturbance caused by the fine stage.

In this paper, an end-point position control strategy is adopted for the dual stage. The y_f measured by a laser interferometer is applied to feedback for controller design of the fine stage in absolute space (i.e., the fine stage tracks the desired trajectory). Preventing saturation of the fine stage is essential for the end-point control strategy. Therefore, the reference trajectory of the coarse stage is designed to track the fine stage (i.e., the trajectory of the coarse stage is y_d-e, where y_d is the reference trajectory of the dual stage in absolute space and e the tracking error of the fine stage).

For an air-bearing table, which provides zero friction, the viscous damping coefficient and coupling spring constant can be ignored. According to the analysis in Section 1, inertia force acting on the fine stage from the coarse stage can be ignored because of end-point control. However, the electromagnetic effect between the two subsystems should be considered for ultra-precise control.

The effect whereby the relative velocity ripple will cause changes in the back electromotive force (back EMF) can be determined using the VCM coil's voltage equation. This is

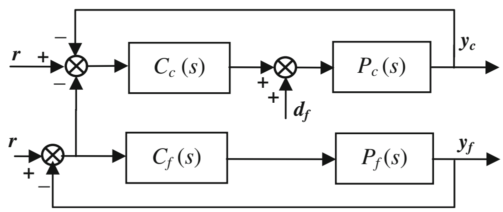


Fig. 2 Block diagram of the dual-stage control system

because the stationary part (stator) of the VCM is mounted on the coarse stage. The circuit diagram of the current amplifier for the VCM driver is shown in Fig. 3, where R_{id} and R_0 are input resistance and output resistance of the current amplifier, respectively; R_{C} and L_{C} are resistance and inductance of the VCM coil, respectively; R_{s} is resistance of current sensor; R_{f} is feedback resistance; u_{i} is current reference outputted by velocity loop; and i is coil current.

Considering $u_0 = e_{\sum} A_{\mathrm{od}}$ and $e_{\sum} << u_{\mathrm{c}}$, the simplified block diagram of the control system's current loop system is shown in Fig. 4, where $E{=}k_{\mathrm{e}}v_{\mathrm{cf}}$ is back EMF, k_{e} is back-EMF constant, and v_{cf} is relative velocity of the two stages.

Generally, $R_{\mathrm{id}}{>>}R_1$, $\frac{R_{\mathrm{id}}}{R_1+R_{\mathrm{id}}} \approx 1$, so we obtain

$$\left(u_{\mathrm{i}} + \frac{R_1 R_{\mathrm{s}}}{R_{\mathrm{f}}} i\right)(-A_{\mathrm{od}}) - k_{\mathrm{e}} v = i(R_{\mathrm{c}} + R_0 + R_{\mathrm{s}} + L_{\mathrm{c}} s)$$

then

$$-A_{\mathrm{od}} u_{\mathrm{i}} - k_{\mathrm{e}} v = i\left(L_{\mathrm{c}} s + R_{\mathrm{c}} + R_0 + R_{\mathrm{s}} + A_{\mathrm{od}} \frac{R_1 R_{\mathrm{s}}}{R_{\mathrm{f}}}\right) \qquad (3)$$

In Eq. 3, if $A_{\mathrm{od}}{>>}k_e$ (i.e., the current closed-loop gain is large enough), then the disturbance caused by back EMF will be obviously attenuated.

Electromagnetic force outputted by the VCM acts on the coarse stage and degrades its control performance because the fine stage is mounted on top of the coarse stage. Unlike the fine stage with a small stroke, by having a simpler mechanical structure and clean response in its working frequency domain, the coarse stage with a heavy load has a complicated mechanical structure and strongly nonlinear problems (e.g., nonlinear force ripple over a large workspace). It is difficult to obtain an accurate model of the coarse stage. Nonetheless, the end-point control strategy does not require high tracking

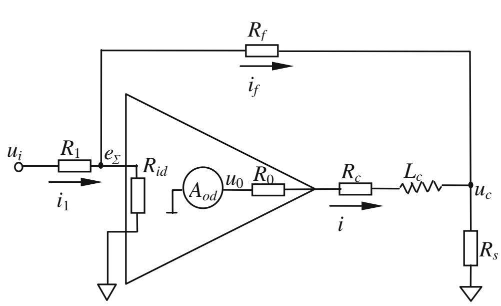


Fig. 3 Simplified circuit for amplifier and coil of VCM

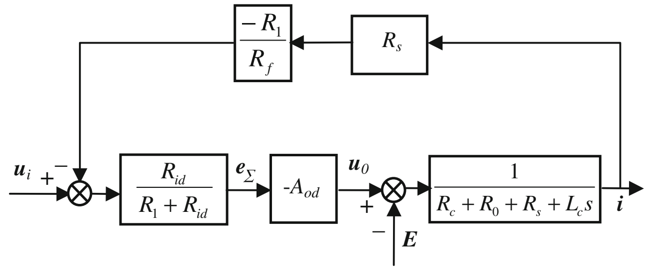


Fig. 4 Block diagram of amplifier and coil of VCM

accuracy of the coarse stage, as the movement range of the VCM is large enough for the tracking capacity of the coarse stage, so a simpler control method can be adopted to avoid modeling the coarse stage. However, the robustness of the coarse-stage controller should be enhanced to prevent instability. In this strategy, high-bandwidth controller design of the fine stage is a key issue for achieving high-speed, ultra-precise linear motion system.

3 Control of the fine stage

3.1 Model identification of the fine stage

If the feasible plant model and its error boundary could be obtained by identification technology, the conservatism will be reduced in a robust controller design. Identifying a model of limited complexity is the first step of the model-based control for an electromechanical servo mechanism.

The open-loop system of the fine stage equipped with zero-friction, air-bearing table is unstable. In practice, closed loop identification is usually adopted for these objects. Closed loop identification can guarantee safety since the object will move in its working range during the identifying process, which can be performed while the stage is working. Hansen and Franklin [5] proposed a framework based on factorization to deal with close-loop identification, in which a system's model is presented through a quotient obtained by two stable factors [i.e., $P{=}N/D$, P is normal model; $(N{,}D)$ are its right coprime factorization]. Consequently, identification for an unstable open-loop object also became identification for two separate stable factors from closed loop data.

In this paper, a method with normalized coprime factorization from closed loop experimental data is used to identify the accurate model of the fine stage.

Normalized coprime factor representation corresponds to systems of the lowest McMillan degree [6]. Consequently, it is imperative that identification within a fractional framework is feasible based on data that correspond to normalized coprime factors of the system. Van den Hof [7] proposed an algorithm for approximate identification of normalized coprime factors.

To initialize the algorithm, an auxiliary model P_x, a data filter F, and an intermediate signal x have to be derived sequentially. P_x is the approximation of plant P_0 and is internally stabilized by the known controller C. It is appropriate to get P_x by an indirect closed loop identification with an output error model structure. With the normalized right coprime factors *nrcf* (N_x, D_x) of the auxiliary model P_x constructed, the data filter is expressed as:

$$F(q) = (D_x(q) + C(q)N_x(q))^{-1} \tag{4}$$

and the intermediate signal x is expressed as:

$$x(t) = F(q)[C(q)I]\begin{bmatrix} y(t) \\ u(t) \end{bmatrix} \tag{5}$$

where $y(t)$ and $u(t)$ are control input of plant and corresponding output, respectively. After the initialization, the algorithm is built up from two steps:

1. The coprime factors of P_0, ($N_{0,\mathrm{F}}$, $D_{0,\mathrm{F}}$), which are accessible from closed loop data, will be shaped in such a way that ($N_{0,\mathrm{F}}$, $D_{0,\mathrm{F}}$) becomes normalized, where ($N_{0,\mathrm{F}}$, $D_{0,\mathrm{F}}$) is a right coprime factor representation. This step is performed as follows:

   1.1. Use the signals $x(t)$ and $[y(t)\ u(t)]^{\mathrm{T}}$ in an identification algorithm with an output error model structure, minimizing $\|\varepsilon(t, \theta)\|_2$ over θ, with

   $$\varepsilon(t,\theta) = \begin{bmatrix} y(t) \\ u(t) \end{bmatrix} - \begin{bmatrix} N(q,\theta) \\ D(q,\theta) \end{bmatrix} x(t)$$

   Considering $[y(t)\ u(t)]^{\mathrm{T}}$ as output and intermediate signal $x(t)$ as input signal, the factorization $[N(q, \theta), D(q, \theta)]$ can be estimated here.

   1.2. Update the auxiliary model $P_x(q)$ on the basis of factorization $[N(q, \theta), D(q, \theta)]$ by

   $$P_x(q,\theta) = N(q,\theta)D(q,\theta)^{-1}$$

   1.3. Again, construct a *nrcf*(N_x, D_x) of the auxiliary model P_x, and according to equations (4) and (5), construct data filter F and intermediate signal x.If the intermediate signal $x(t)$ and the signal $[y(t)\ u(t)]^{\mathrm{T}}$ are normalized, the second step of the algorithm will be invoked; otherwise, repeat steps 1.1 to 1.3.

2. With intermediate signals x(t) and $[y(t)\ u(t)]^{\mathrm{T}}$ from the last step, an approximate identification of the normalized factorization ($N_{0,\mathrm{F}}$, $D_{0,\mathrm{F}}$) using an output error structure as:

$$\begin{bmatrix} N(q,\theta) \\ D(q,\theta) \end{bmatrix} = \begin{bmatrix} B_{\mathrm{N}}(q,\theta) \\ B_{\mathrm{D}}(q,\theta) \end{bmatrix} A(q,\theta)^{-1} \tag{6}$$

To perform identification with normalized coprime factors, a PD controller is designed to stabilize the plant. In order to excite the plant sufficiently, multi-sinusoids are chosen as the excitation signal $r(t)$:

$$r(t) = \sum_{k\in\{0,\ldots,N-1\}} a_k \cos(\omega_k t + \varphi_k)$$

where ω_k are from 1 Hz to 5 kHz with the interval of 1 Hz, a_k are the magnitudes of 10 nm for all frequencies, and $\varphi_k \in [0, 2\pi]$ are random variables with normal distribution.

Applying the algorithm described above, an almost normalized coprime factorization can be derived after repeating the first step of the algorithm twice. With the approximate identification proposed by the second step, a model with an 18th order is attained. The pole-zero pairs are given in Fig. 5.

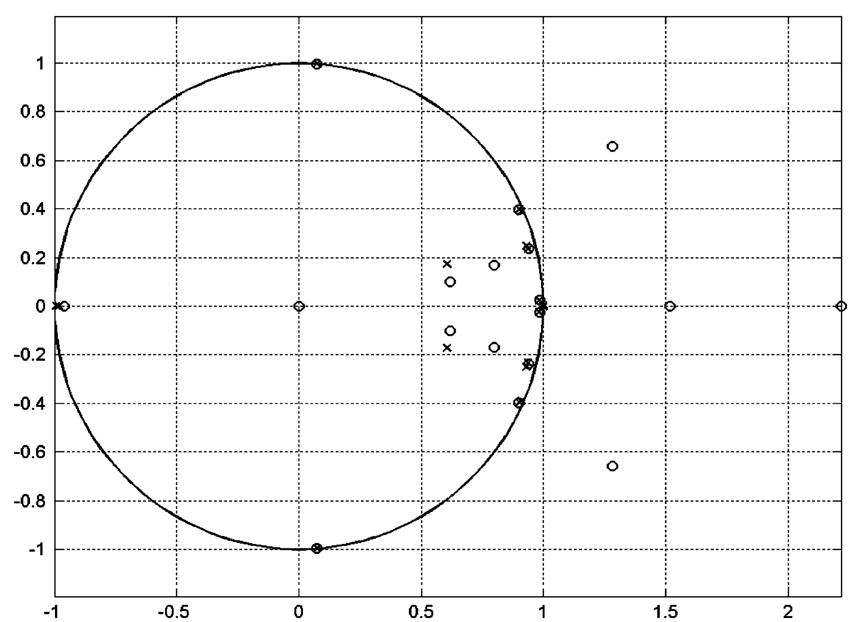


Fig. 5 The poles (*ex symbol*) and zeros (*open circle*) of the model

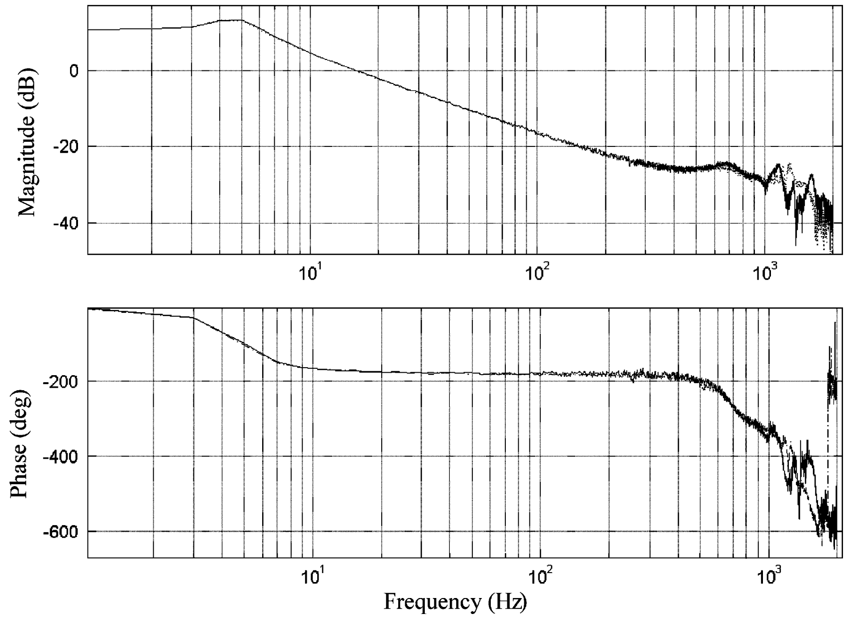


Fig. 6 Identification under different coarse-stage states

3.2 Model reduction

A relatively high 18th order model is derived from the identification above. Therefore, mode reduction has to be performed. In this experiment, model reduction is performed as follows:

1. Approximate pole-zero cancelation

   The polynomial-based algorithm suffers from a numerical problem, such that the pole-zero pairs that should be canceled are not equivalent. Thus, approximate pole-zero cancelation has to be performed manually for minimal realization. This type of cancelation can be used in $P_x(q)$ and can prevent the data filter F from being unstable because of numerical faults.

2. Balanced reduction

   For a reachable, observable, and stable system, a balanced realization and a diagonal controllability and observability grarnmian can be produced because the grarnmian reflects the combined controllability and observability of individual states of the balanced model [8]. The states with a small grarnmian are those that are less controllable and observable. Therefore, those states can be deleted while retaining the most important input–output characteristics of the original system.

   For an unstable system P_0, the balanced reduction can be done only in an unstable part. Thus, P_0 should be decomposed into stable part P_s and unstable part P_u, where

$$p_0 = \frac{B_0}{A_0} \quad p_s = \frac{B_s}{A_s} \quad p_u = \frac{B_u}{A_u} \quad A_0 = A_s A_u$$

   and B_s and B_u can be derived from the Diophantine equation as follows:

$$A_s B_u + A_u B_s = B_0$$

3. Cancel the modes in high-frequency regions

   The modes in high-frequency regions are from real modes in plant and sampling aliasing. However, those modes in frequency regions ten times more than the desired bandwidth are negligible in control design.

After model reduction, we applied the following steps: An identified sixth order model of the fine stage with a 5-kg load is as follows:

$$P_x = \frac{0.00140z^6 - 0.01055z^5 + 0.03343z^4 - 0.05694z^3 + 0.05468z^2 - 0.02787z + 0.00587}{z^6 - 5.67929z^5 + 13.64778z^4 - 17.76353z^3 + 13.20752z^2 - 5.31911z + 0.90662}$$

By model identification, the aforementioned analysis of back EMF disturbance in Section 2.2 can be validated. Using a closed current loop, with the position and velocity loops opened, the model identification of the fine stage is performed in the frequency domain. In Fig. 6, the solid line presents the identified model of the fine stage, the coarse

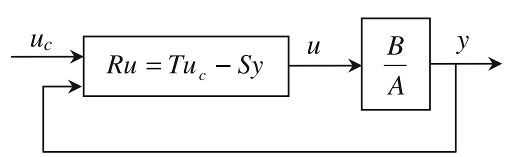


Fig. 7 Structure of the fine-stage controller

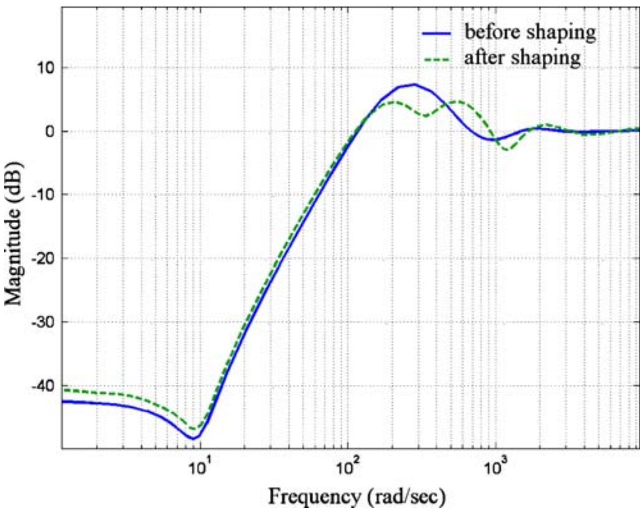


Fig. 8 Sensitivity before and after sensitivity function shaping

stage is locked, the dashed line represents the identified model of the fine stage, and the coarse stage is moving. With the locked coarse stage, the identification result of every experiment may differ in the frequency range, exceeding 1 kHz. In Fig. 6, the initial condition and noise bring about a difference in the two experimental curves at a frequency range higher than 1 kHz. The bode plot proves that the coupled effect can be ignored for the fine-stage controller design of the air-bearing dual stage.

3.3 Control of the fine stage

A robust polynomial pole placement with sensitivity function shaping is used in the control design. The fine-stage controller's structure is shown in Fig. 7, where u_c is the desired command, u is the output of the controller, and y is the output.

Initially, a full order (sixth order) controller is designed with the dominant pole at 200 Hz and a relative damp ratio of 0.707. The characteristic polynomial A_{cl} can be factorized as follows:

$$A_{cl} = \mathrm{Ac}_1 \cdot \mathrm{Ac}_2 \cdot \mathrm{Ac}_3 \cdot \mathrm{Ao}_1 \cdot \mathrm{Ao}_2 \cdot \mathrm{Ao}_3$$

where Ac_1 corresponds to the dominant pole, Ac_2 and Ac_3 correspond to the auxiliary poles at 1,600 Hz and a relative damp ratio of 0.707, Ao_1 corresponds to the observer for the dominant pole, with its roots lain at 400 Hz and a relative damp ratio of 0.707; and Ao_2 and Ao_3 correspond to the notch filters of the resonance mode.

Solving the Diophantine equation AR+BS=A_{cl}, R, S, and T of the controller can be derived as follows:

$$R = z^6 - 4.368z^5 + 8.066z^4 - 8.081z^3 + 4.654z^2 - 1.471z + 0.2013$$

$$S = 2.809z^5 - 13.08z^4 + 24.95z^3 - 24.35z^2 + 12.16z - 2.486$$

$$T = 4.911z^6 - 27.03z^5 + 63.06z^4 - 79.76z^3 + 57.65z^2 - 22.56z + 3.734$$

However, the controller designed above has a maximum sensitivity function of 7.38 dB, and a typical robust controller requiring the maximum sensitivity function must be less than 6 dB [9]. According to the method in [9], a sensitivity function shaping factor R' is added to R.

$$R'_1 = \mathrm{z}^2 - 1.928\mathrm{z} + 0.9389$$

With the sensitivity function shaping factor R', R, S, and T are redesigned as follows:

$$R = z^8 - 5.866z^7 + 15.09z^6 - 22.32z^5 + 20.87z^4 - 12.77z^3 + 5.066z^2 - 1.217z + 0.138$$

$$S = 10.51z^7 - 69.5z^6 + 199.2z^5 - 320.8z^4 + 313.6z^3 - 186.1z^2 + 62.04z - 8.966$$

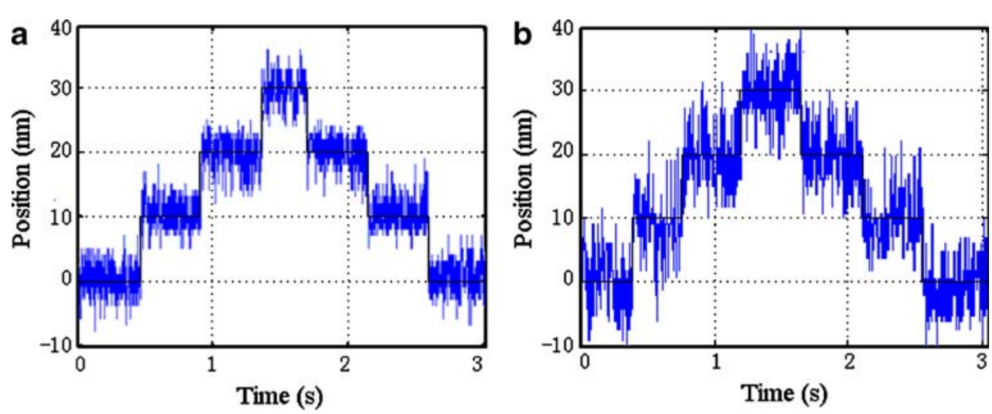
$$T = 0.1503z^6 - 0.8272z^5 + 1.93z^4 - 2.441z^3 + 1.764z^2 - 0.6905z + 0.1143$$

The new controller has a maximum sensitivity function of 4.7 dB. Figure 8 shows the closed loop sensitivity functions before and after sensitivity shaping.

3.4 Experimental results of the fine-stage control system

Using the abovementioned methods, the high-precision robust controller for the fine stage is obtained. The closed-loop bandwidth of the fine stage with a 5-kg load

Fig. 9 Comparison of continuous step response of 10 nm



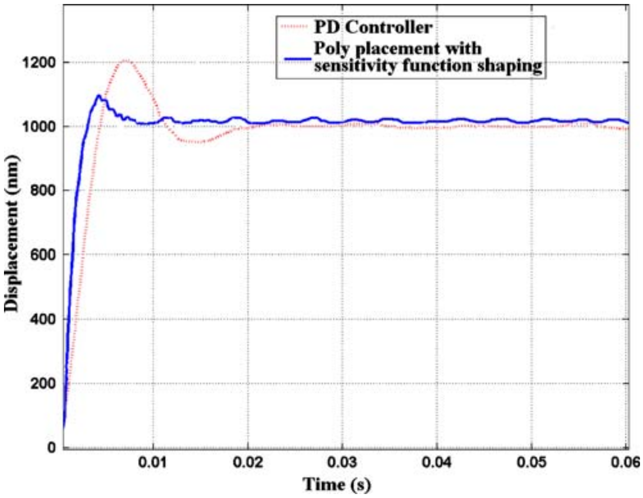


Fig. 10 Comparison of step response of 1 μm

is more than 200 Hz. In this paper, Fig. 9a is the continuous step response of 10 nm using a controller, and Fig. 9b shows the continuous step response of 10 nm using a PD controller. The black line is the demanded continuous step. Figure 10 compares the two controllers through a step response of 1 μs. Figures 9a and b and 10 show that the controller designed by poly placement with sensitivity function shaping can achieve a faster response and smaller positioning waviness.

4 Experimental results of the dual stage

The OS of the control computer is RTLinux. The industrial computer works with dual CPUs. By appropriate scheduling with RTlinux, one of the two CPUs responds with a real-time interrupt task and the other deals with a non-real-time calculation, such as a draw interface. The servo period of the fine stage is 50 μs. The coarse stage controller is a control board based on digital signal processing, which communicates with the control computer through a PCI bus. The servo period of the coarse stage is 110 μs. The controller of the PMSLM is designed by a proportional–

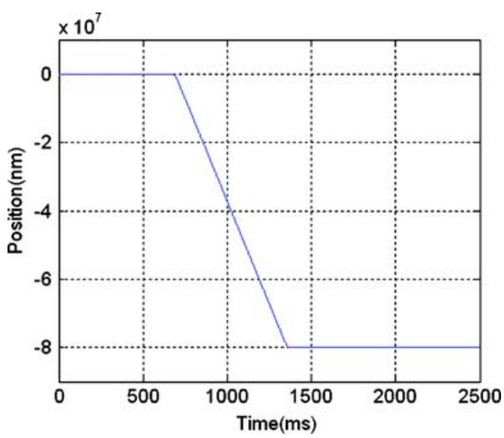


Fig. 11 The planning trajectory

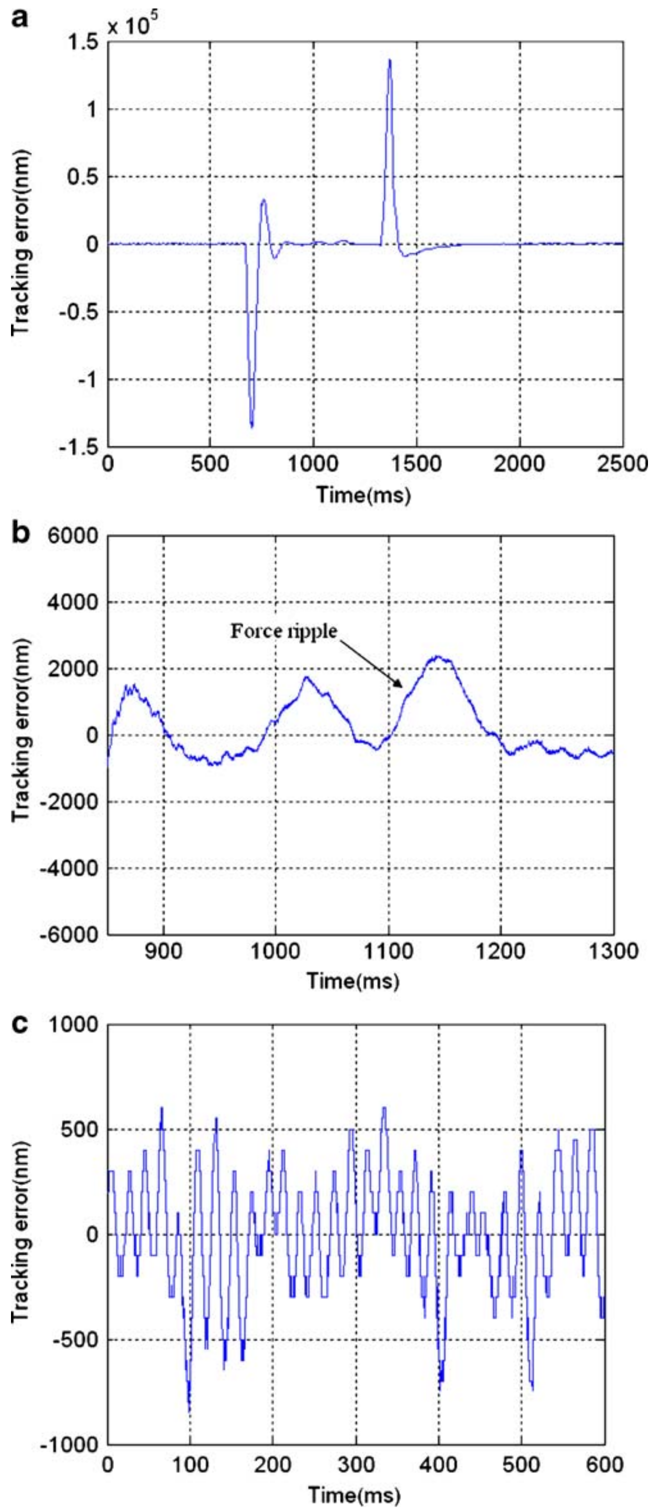


Fig. 12 Tracking error of the coarse stage with 5-kg load

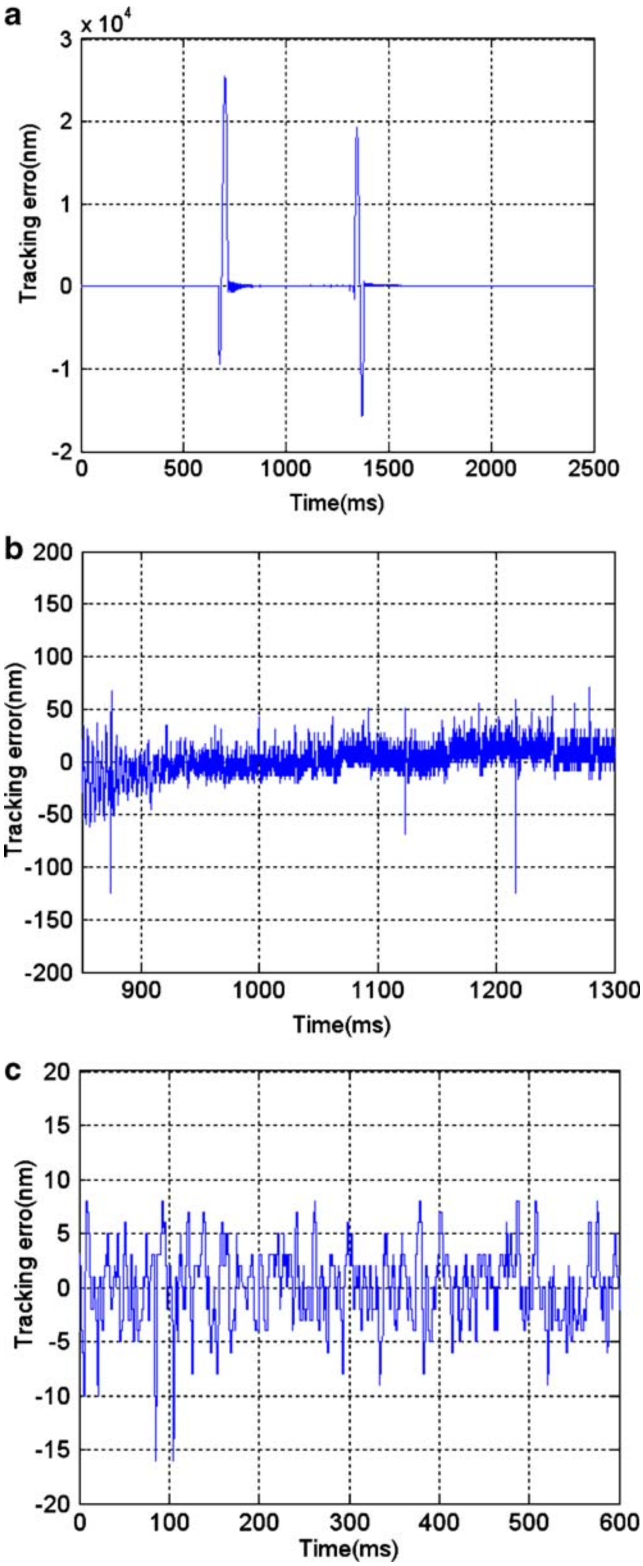


Fig. 13 Tracking error of the fine stage with 5-kg load

integral–derivative algorithm. Acceleration feed-forward is added to the coarse stage controller for a rapid response during acceleration/deceleration. The real-time position is measured by the dual-frequency laser interferometer.

Experiment 1 The load of the dual stage is 5 kg. Desired velocity and acceleration are 120 mm/s and 0.4 g, respectively.

Structural vibration excited by environment and motion limits the stage's servo performance in ultra-precision control. For isolating the environmental vibration, the system uses four air springs as passive vibration isolation elements. On the other hand, for reducing the vibration caused by motion with a heavy load, especially during acceleration/deceleration, the S-curve acceleration/deceleration for the dual-stage trajectory planning is adopted. In this algorithm, the jerk of the motion is continuous, which is useful for weakening the vibration during acceleration/deceleration.

In this paper, the S curve for trajectory planning is fitted by a five-degree polynomial as follows:

$$y_{\mathrm{d}}(kT) = A_{\mathrm{r}}\left[6\left(\frac{kT}{T_{\mathrm{r}}}\right)^5 - 15\left(\frac{kT}{T_{\mathrm{r}}}\right)^4 + 10\left(\frac{kT}{T_{\mathrm{r}}}\right)^3\right] \tag{8}$$

where A_{r} is accelerating distance, T_{r} is accelerating time, T is sample period of the control system, and k is number of the sample period. For a demanded maximum acceleration $a_{\max}$ and velocity $v_{\max}$ of the system, using Eq. 8, we obtain:

$$A_{\mathrm{r}} = 1.6\frac{V_{\max}^2}{a_{\max}} \text{ and } T_{\mathrm{r}} = 3\frac{V_{\max}}{a_{\max}}$$

Fig. 11 presents the planning trajectory.

Figures 12 and 13 show the tracking errors of the dual stage. Figure 12a presents the tracking error of the coarse stage. The steady-state tracking error and positioning error are shown, respectively, in Fig. 12b, and c, which are enlarged images of Fig. 12a. The positioning accuracy (peak-to-peak value) of the coarse stage is about ±0.5 μm, with a tracking error of approximately 2 μm. The nonlinear force ripple of the coarse stage is shown in Fig. 12b. Using

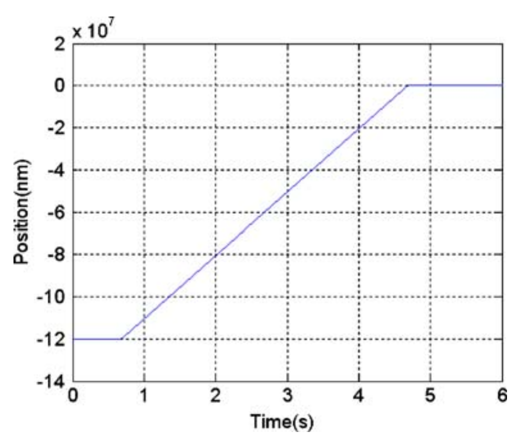


Fig. 14 Reference trajectory of the dual stage with 20-kg load

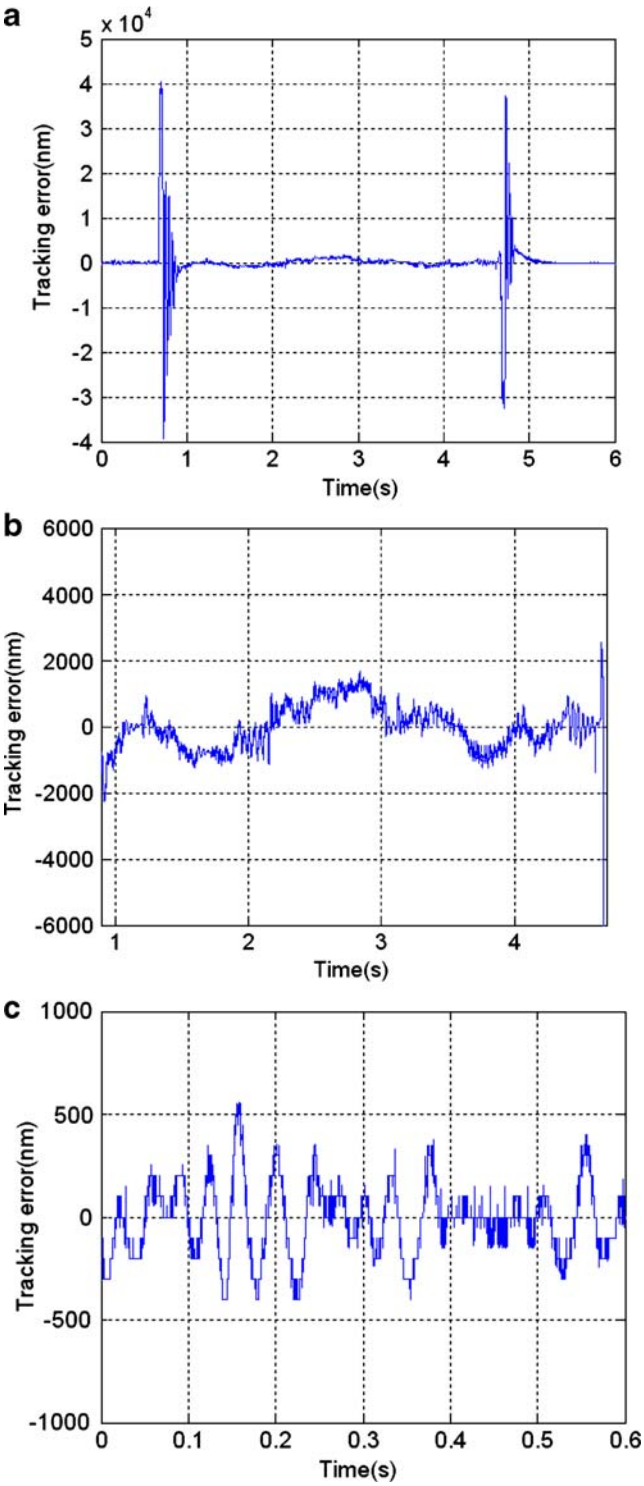


Fig. 15 Tracking error of the coarse stage with 20-kg load

some control methods such as iterative learning control, the ripple can be weakened and the tracking performance of the coarse stage can be improved. However, the stability margin may be reduced due to the high control gain and disturbances from the fine stage, which tends to cause instability in the coarse stage while the fine stage moves. According to the analysis in Section 2.2, a highly precise coarse stage is not required, in that the fine stage has a long stroke and better tracking performance. We tune the parameters of the PID controller to guarantee that the

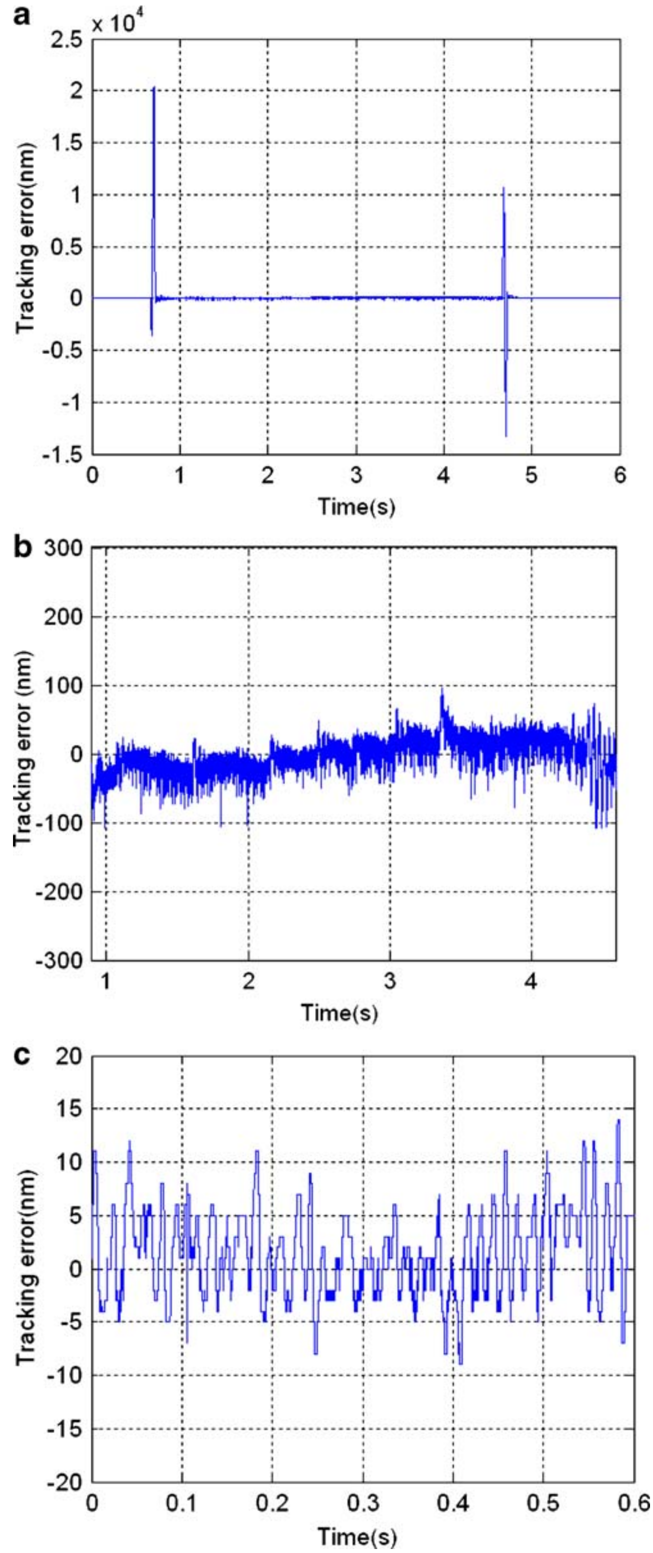


Fig. 16 Tracking error of the fine stage with 20 kg load

tracking error of the coarse stage is less than the workspace of the fine stage. It is easy to implement because the stroke of the fine stage is 10 mm.

Figure 13a shows the tracking error of the fine stage. The steady-state tracking and positioning errors are shown in Figs. 13b and c, respectively; these figures are enlarged images of Fig. 13a. Positioning and tracking accuracy is greatly improved because of the fine stage's high bandwidth. The positioning accuracy (peak-to-peak value) is about 10 nm, which is same as the experimental results in Section 3.4, while the coarse stage is locked. Disturbances in the high frequency caused by the electromagnetic effect from the positioning control of the coarse stage are attenuated by the controller of the fine stage. In Fig 13b, the tracking accuracy is less than 50 nm. The designed controller do not compensate for the force ripple from the coarse stage, however, the negative effect of the force ripple cannot be observed in Fig 13b. These prove that the coupling effect from the coarse stage can be ignored in the controller design of the fine stage. The experimental results are consistent with the analysis in Section 2.2 and in Fig. 6 of Section 3.2.

Compared to the single stage driven by PMLSM, the use of the fine-stage positioning and tracking errors decreased about 100 times and 40 times, respectively.

Experiment 2 The load of the dual stage is 20 kg. Desired velocity and acceleration are 30 mm/s and 0.1 g, respectively. The reference trajectory, which is also calculated by Eq. 8, is shown in Fig. 14. The interpretations for Figs. 15 and 16 correspond to those for Figs. 12 and 13.

We also applied the methods in Section 3 for controller design of the fine stage with 20-kg load. The closed loop bandwidth with 20-kg load is 80 Hz.

With 20 kg of the load, the stage exhibits a slower response speed and requires a longer adjustment period. Comparing Figs. 15b and 12b, the force ripple in Fig. 15b is less than that in Fig. 12b because of the stage's slower moving velocity. With different loads, the positioning accuracy of the dual stage is similar. However, the tracking capacity of the fine stage with heavier load became worse. The peak-to-peak value of the steady-state tracking error with 5-kg load is about half as much as that of the stage with 20 kg.

5 Conclusion

In this paper, an air-bearing dual-stage control was presented that allowed for the implementation of high-speed, ultra-precise linear motion. We showed that, due to the air-bearing mechanical structure of such a system, the controller for the fine stage could be independently designed. Through closed-loop identification based on normalized coprime factors, mode reduction, and polynomial pole placement with sensitivity function shaping, the motion control of the fine stage was performed. The coarse stage tracked the trajectory of the fine stage to prevent fine-stage saturation. Moreover, a high-precision control was not required as long as control stability was guaranteed. The dual-stage control performance was determined by the end-point control strategy of the fine stage. Experimental results proved that this scheme could be achieved with satisfactory control performance.

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